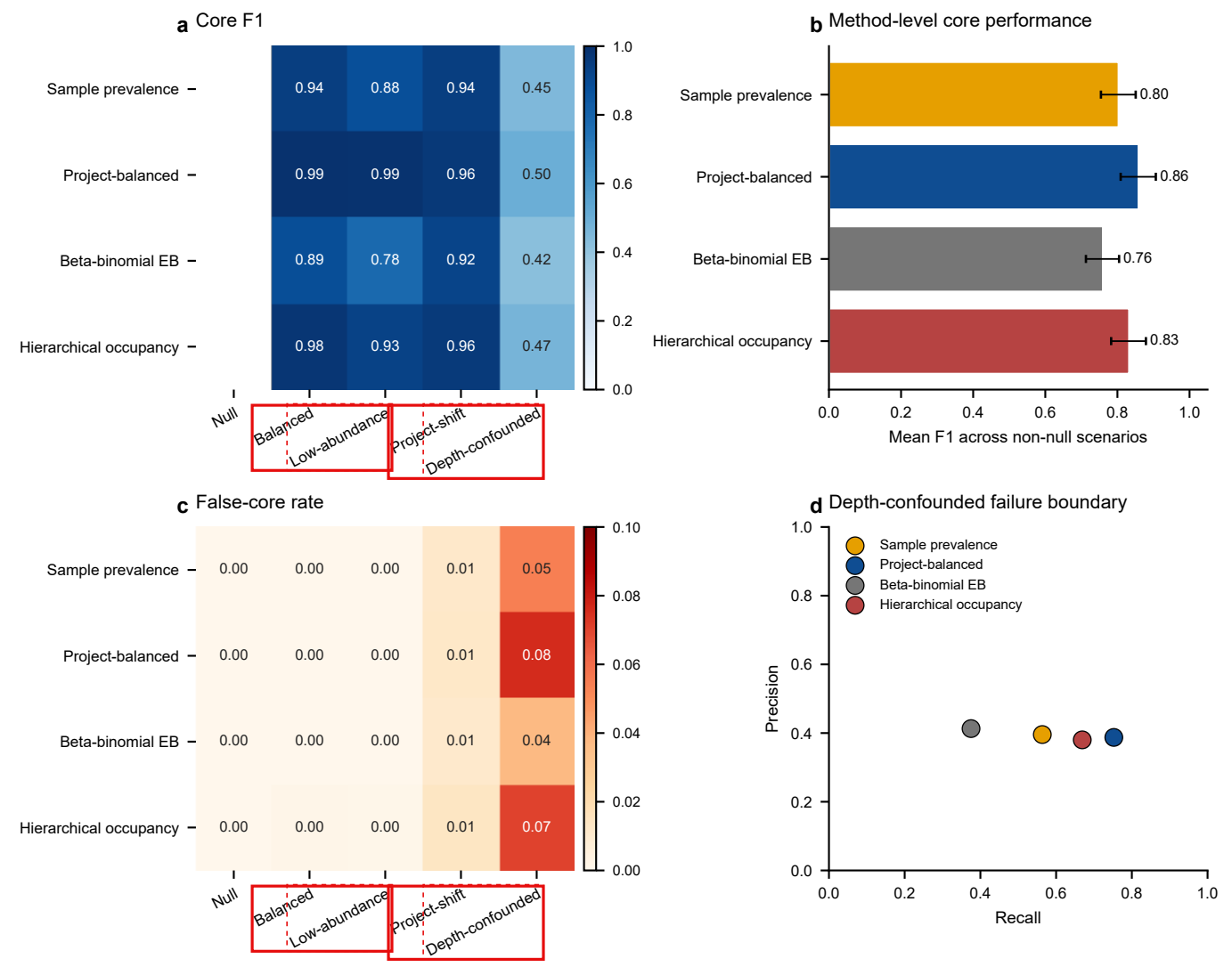
a Core F1
Sample prevalence
Project-balanced
Beta-binomial EB
Hierarchical occupancy
0.94 0.88 0.94 0.45
0.99 0.99 0.96 0.50
0.89 0.78 0.92 0.42
0.98 0.93 0.96 0.47
1.0 0.8 0.6 0.4 0.2 0.0
Null
Balanced
Low-abundance
Project-shift
Depth-confounded
b Method-level core performance
Sample prevalence
Project-balanced
Beta-binomial EB
Hierarchical occupancy
0.80
0.86
0.76
0.83
0.0 0.2 0.4 0.6 0.8 1.0
Mean F1 across non-null scenarios
c False-core rate
Sample prevalence
Project-balanced
Beta-binomial EB
Hierarchical occupancy
0.00 0.00 0.00 0.01 0.05
0.00 0.00 0.00 0.01 0.08
0.00 0.00 0.00 0.01 0.04
0.00 0.00 0.00 0.01 0.07
0.10 0.08 0.06 0.04 0.02 0.00
Null
Balanced
Low-abundance
Project-shift
Depth-confounded
d Depth-confounded failure boundary
Sample prevalence
Project-balanced
Beta-binomial EB
Hierarchical occupancy
Precision
Recall
0.0 0.2 0.4 0.6 0.8 1.0